Answer **all** the questions in the spaces provided.

1 **(a)** State what is meant by the *moment of a force*.

.....

.....

..... [1]

(b) A traffic light hangs from a uniform metal pole AB, freely pivoted at point A, as shown in Fig. 1.1. The pole is 6.5 m long and has a mass of 9.0 kg. The mass of the traffic light is 11.0 kg.

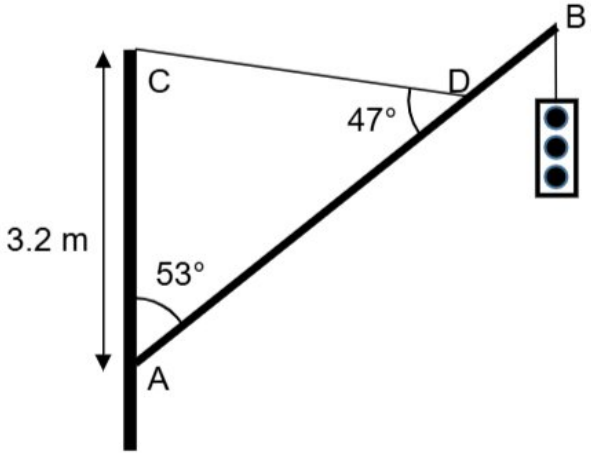


Fig. 1.1

(i) Show that the tension in the cable CD is 250 N.

[2]

(ii) Determine the magnitude of the force exerted on the metal pole at point A.

force = ... N [2]

(iii) Explain how the magnitude of the force at the point A changes if the traffic light is shifted from B to D.

.....

.....

.....

..... [2]

[Total: 7]